

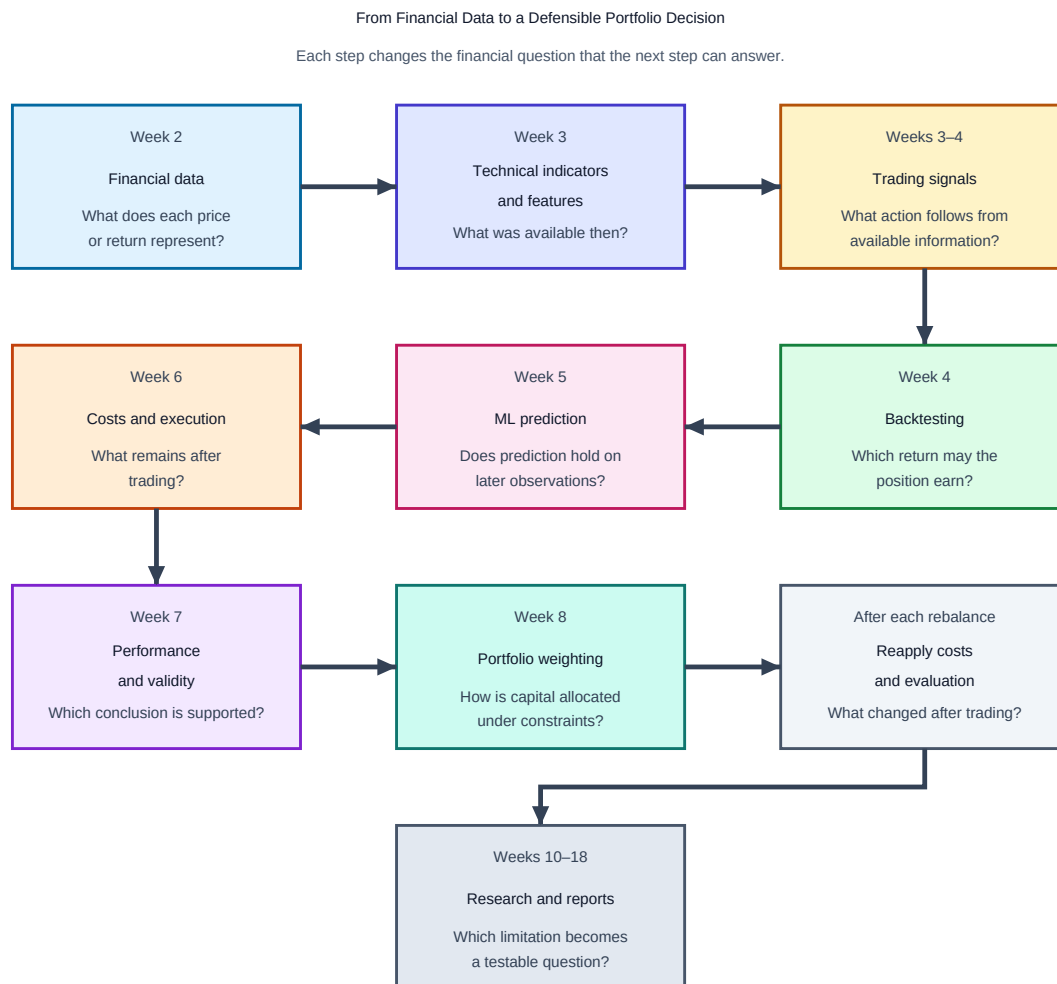
Week 1 - From an Investment Idea to a Testable Quantitative Trading Question

Read the complete course syllabus.

What you will learn

Before building a trading strategy, decide what result would answer your investment question. This week, you will turn one broad idea into a question with a stated decision, comparison, and outcome. No coding or return calculation is required.

Course workflow



Every result must preserve data, time, units, assumptions, and common comparison conditions.

Figure 1. The course builds a trading strategy step by step. Changing portfolio weights requires checking costs and performance again. The workflow is reproduced above.

1. Before following the workflow, decide what you want to test

Figure 1 shows what we will build. It does not tell us **what would count as evidence for our investment idea**. That decision comes first: it determines which data, model, backtest, and comparison we need.

Consider this idea:

AI can select better investments.

“Better” could mean a higher return, a smaller loss, or less trading. These are different questions. For this example, choose one: **does the model-based allocation earn a higher return after costs than a fixed allocation?**

We will specify this comparison before calculating anything.

2. Turn the idea into one comparison

The following settings are illustrative, not the official Report 1 requirements. They describe a proposed experiment, not observed investment results.

Decision to specify	Settings for this example
Assets and baseline	Apple common stock (intended label AAPL), Microsoft common stock (MSFT), and USD cash. Compare the model-based allocation with 50% in each stock.
Model-based allocation	Predict each stock's next-session opening-to-closing price return. Set negative predictions to zero and allocate in proportion to the positive predictions. If neither is positive, hold cash. No short selling.
Information and decision time	Use a session's daily prices and volume only after they have actually been received. Finish the prediction and allocation before the next regular session opens; a closing timestamp alone does not establish availability.
Execution and holding interval	Both portfolios assume stock purchases at the next regular session's opening and sales at its closing. These are execution assumptions, not verified fills.
Development and evaluation	Fit and select the model and rule using outcomes ending no later than December 29, 2023. Evaluate opening-to-closing intervals from January 2 through December 31, 2024, without using those outcomes for fitting or selection.
Common conditions	Same securities, evaluation sessions, starting capital of USD 100,000, execution schedule, and cost assumptions. Charge 10 basis points (0.10%) of each amount bought or sold.
Primary outcome	Net cumulative return: the percentage change in portfolio value over the evaluation period after costs. A result no higher than the baseline would not support the proposed advantage.

Before running this experiment, verify the securities, price fields, and trading calendar. Also specify whether cash earns interest and how trading costs are deducted from available funds. Allocations must be affordable after costs. Apply these conventions to both portfolios; the assumed cost is not a measured historical fee.

Now the question can be short because the table supplies its conditions:

Under these settings, does the model-based allocation produce a higher net cumulative return than the 50%-50% baseline during 2024?

This compares the **complete allocation rules**, not the prediction model alone. Even a higher historical net return would not establish future profitability. At this stage, we have a question and proposed settings-not an answer or a fully implemented backtest.

3. Check whether the comparison answers that question

Suppose Team A reports a return before costs, while Team B reports a return after costs. Their numbers do not answer our question on common terms. Both must apply the same cost definition before comparing net returns.

Timing matters too. If a day's price data arrive at 16:05, a decision at 16:00 cannot use them. This is a hypothetical clock example, not a claim about a provider's release schedule. Moving the decision later still requires an execution opportunity after the decision.

These checks explain why Figure 1 starts with data and proceeds through signals, execution, and evaluation. A sophisticated model cannot repair unavailable inputs or inconsistent comparisons.

4. Your turn: make one change and rewrite the question

Keep the example's assets and evaluation period. Choose a different allocation rule to compare with the same baseline; describe the rule in words without calculating weights or returns.

Use one short table:

Question to settle	Your proposed answer
What allocation rule changes, and what baseline remains?	
Which inputs are needed, when are they available, and when can the decision be executed?	
Which outcome and holding interval will be compared?	
Which data, dates, costs, and other conditions stay fixed?	
What result would not support your proposed advantage?	
What still needs verification before calculation, and where must the conclusion stop?	

Write one question using your table. Ask a classmate to identify one missing condition that could change the comparison, then revise it. If they find none, ask them to explain why the stated comparison is consistent.

Keep only the table, your question, and the peer comment with your response. The activity is complete when another student can tell what changes, what stays fixed, what evidence would answer the question, and what remains unknown.

Next: obtain the evidence

Week 1 defines the question. [Week 2 - Financial Data and Returns](#) checks what the proposed prices, dates, and returns mean. Week 3 determines which features and signals can be available before a decision. Later weeks build and evaluate the strategy.

The [syllabus](#) and course statement and GenAI policy contain the course rules. This activity does not add a report requirement or a grading rule.

The complete syllabus and course policy follow as appendices. Report instructions are supplied separately; withdrawn source-file links are not active in this PDF.

Appendix A - Course syllabus

115學年度第1學期課程教學大綱

學年：115學年

學期：第1學期

開課班級：M0L20A 資訊管理系碩士班二年甲班

科目代碼：M0L23910

科目名稱：智慧金融科技 (ARTIFICIAL INTELLIGENCE IN FINTECH)

學分數：3.0學分

每週上課時數：3.0小時

任課老師：李文毅

教學目標

This EMI course develops a reproducible workflow for AI-driven quantitative trading and portfolio weighting. The learning sequence is financial data, technical indicators and features, signals, backtesting, machine-learning prediction, trading costs, performance evaluation, and weighting.

After completing the course, students should be able to prepare financial data, preserve information boundaries, construct and evaluate a trading strategy, compare portfolio-weighting methods, develop a clearly defined modification of an existing method, and communicate evidence and limitations in professional English. Papers for Weeks 11, 12, 13, 15, and 18 will be announced later.

教學內容

Students begin with prices, returns, trading calendars, information timing, technical indicators, features, and signals. Every calculation must use information available at the stated decision time.

Students then convert signals into positions and returns, estimate prediction models chronologically, add execution assumptions and transaction costs, and evaluate performance, risk, uncertainty, and backtest validity.

The course next introduces basic portfolio-weighting methods. Through reports and paper discussions, each team develops and evaluates a modification of an existing method under common data, constraints, costs, and out-of-sample conditions.

Course coverage and evidence

Learning area	Weeks	Question students should answer	Observable evidence
Financial question	1	What decision, available information, comparison, and evidence would make an investment idea testable?	A revised quantitative trading question that states the conditions held fixed, a result that would not support the idea, and the remaining limitations.
Financial data and returns	2	What do the asset, field, date, unit, currency, and return represent?	Audited data and return calculations using valid holding intervals.
Features and signals	3	How can a financial idea become a feature and signal without using unavailable information?	A financial hypothesis, trailing features, leakage checks, and a dated signal.
Backtesting	4	How do signals become lagged positions, trades, gross returns, and wealth?	A reproducible gross-return ledger with accounting checks.
Machine-learning prediction	5	How can a model predict a later outcome using only information available at the prediction time?	A dated prediction table, chronological baselines, updating rules, and a protected evaluation period.
Trading costs and execution	6	How do execution and cost assumptions change gross results into net results?	A dated turnover and cost ledger, execution-delay checks, and compatible benchmarks.
Performance and validity	7	What do return, risk, drawdown, uncertainty, and backtest checks permit us to conclude?	Common-condition performance evidence and a conclusion limited to the completed checks.
Portfolio weighting	8	How are dated inputs converted into feasible weights and compared under common conditions?	Constraint checks, target and holding weights, rebalancing, turnover, net results, and sensitivity analysis.
Report 1	10	Can the team connect data, prediction, signals, backtesting, costs, performance, and an initial weighting approach?	A frozen Git version with a team-defined and justified mapping from model output or signals to weights.
Weighting-method research	11-13 and 15	How can a limitation of an established method motivate a precise and testable modification?	Source-based method analysis and development of a proposed modification. Specific papers will be announced later.
Report 2	14	Can the team state its proposed weighting-method modification and present preliminary evidence?	A frozen Git version that connects the cited method, identified limitation, proposed change, and preliminary comparison.
Final evaluation	16-17	Does the modification remain defensible under common out-of-sample conditions?	A frozen final report, oral defense, and evidence-based comparison of the other teams.

Learning area	Weeks	Question students should answer	Observable evidence
Further paper discussion	18	Which unresolved limitation could guide later research?	A paper-based discussion connected to limitations found in the final reports.

Week 8 provides common baseline weighting and comparison procedures. The selected course papers and each team's cited sources must provide any additional mathematical, optimization, or empirical background required for the proposed modification.

第一週

English: Week 1 shows how to convert a general investment idea into a testable quantitative trading question by specifying the decision, available information, outcome, baseline, conditions held fixed, evidence, and limitations.

中文：第一週說明如何明確定義決策、資訊範圍、結果、比較基準、固定條件、證據與限制，將一般投資想法改寫成可檢驗的量化金融問題。

第二週

English: Week 2 covers financial prices, returns, corporate actions, missing values, trading calendars, and variable alignment.

中文：第二週介紹金融價格、報酬率、公司行動、缺失值、交易日曆及變數對齊。

第三週

English: Week 3 covers information timing, technical indicators, financially motivated features, leakage checks, and trading signals.

中文：第三週介紹資訊時間、技術指標、財務特徵、資訊洩漏檢查及交易訊號。

第四週

English: Week 4 converts signals into positions, trades, returns, and a reproducible backtest.

中文：第四週將交易訊號轉換為部位、交易與報酬，並建立可重現的回測。

第五週

English: Week 5 introduces machine-learning prediction, chronological evaluation, rolling and expanding windows, and walk-forward updating.

中文：第五週介紹機器學習預測、時間順序評估、滾動與擴展視窗，以及逐步向前更新。

第六週

English: Week 6 adds transaction costs, execution delay, turnover, and benchmark comparisons to backtests.

中文：第六週將交易成本、執行延遲、週轉率及基準比較納入回測。

第七週

English: Week 7 evaluates return, risk, drawdown, turnover, uncertainty, overfitting, and backtest validity.

中文：第七週評估報酬、風險、回撤、週轉率、不確定性、過度配適及回測有效性。

第八週

English: Week 8 introduces portfolio constraints and equal, market-capitalization, signal-based, and inverse-volatility weighting.

中文：第八週介紹投資組合限制、等權重、市值加權、訊號式加權及反波動度加權。

第九週

English: Week 9 has no class because of instructor travel.

中文：第九週因教師出國停課。

第十週

English: Week 10 contains Report 1 on prediction, quantitative trading, backtesting, and the initial weighting approach.

中文：第十週進行第一次報告，內容包含預測、量化交易、回測及初步加權方法。

第十一週

English: Week 11 discusses papers related to weighting methods.

中文：第十一週討論與投資組合加權方法相關的論文。

第十二週

English: Week 12 discusses papers related to weighting methods.

中文：第十二週討論與投資組合加權方法相關的論文。

第十三週

English: Week 13 discusses papers related to weighting methods.

中文：第十三週討論與投資組合加權方法相關的論文。

第十四週

English: Week 14 contains Report 2 on the proposed weighting-method modification and preliminary evidence.

中文：第十四週進行第二次報告，說明加權方法的修改與初步證據。

第十五週

English: Week 15 discusses papers related to weighting methods.

中文：第十五週討論與投資組合加權方法相關的論文。

第十六週

English: Week 16 is the first final-report session.

中文：第十六週進行第一場期末報告。

第十七週

English: Week 17 is the second final-report session, followed by evidence-based comparison and ranking of the other teams.

中文：第十七週進行第二場期末報告，並以證據比較與排序其他組別。

第十八週

English: Week 18 discusses papers related to weighting methods.

中文：第十八週討論與投資組合加權方法相關的論文。

成績評量

- Class participation and weekly learning evidence - 40 points
- Report 1 - Quantitative Trading Report: 15 points
- Report 2 - Weighting Method Progress Report: 15 points
- Report 3 - Final Report: 30 points

Detailed requirements for the three reports will be provided in the corresponding report-week files.

課程運作與繳交說明

The course statement and GenAI policy apply throughout the semester. Detailed report instructions are provided for Report 1, Report 2, and the final report.

Report submission through GitHub

Reports are submitted through GitHub according to the instructions in the corresponding report-week file. The main report, presentation, and oral explanation use English. Do not place credentials, restricted data, grading records, or personal information in a public repository.

Frozen version

At each report deadline, each team identifies one Git commit as the frozen version for presentation and evaluation. Later commits do not replace the frozen version. If an error is corrected afterward, the team must retain the original version and explain how the correction affects the results and conclusions.

Presentation and comparison

Students do not evaluate their own team. Comparisons of other teams must use common financial evidence and clearly stated criteria. Rankings received from classmates do not directly determine team grades. Individual reviews, rankings, and other grading-related personal records must remain private.

Approved accommodations follow the university's official procedure while preserving the assessed learning outcomes.

指定教科書及參考書籍

The course does not require a single textbook. Weeks 2-8 primarily use instructor-prepared notebooks and open or authorized financial datasets. Weeks 11, 12, 13, 15, and 18 use papers related to weighting methods. Assigned papers and accessible versions will be announced through the course GitHub repository.

The course uses each source only for claims it can support:

Source	Used to support	Does not by itself establish
Academic books and papers	Financial and statistical definitions, methods, and research findings	The behavior of a current software package or the accuracy of a particular downloaded observation
Regulator, exchange, company, and data-provider records	Security identity, corporate events, trading calendars, field definitions, and stated observations	That a trading or weighting method is effective
Official software documentation	Package, function, and API behavior	A financial theory or empirical investment claim
Controlled teaching examples	A calculation, assumption, or failure case	A real-market fact or future investment performance

Course materials

The public student materials are organized in the [course directory](#). Begin with the [Week 1 lesson](#), which introduces the course workflow and asks students to turn an investment idea into a testable quantitative trading question.

Instructor answers, grading records, administrative files, personal EMI documents, and material without verified distribution rights are not published in this repository.

Appendix B - Course statement and GenAI policy

Artificial Intelligence in FinTech - Course Statement and GenAI Policy

English version

Educational-use statement

All notebooks, examples, strategies, backtests, and discussions in this course are for educational purposes. They are not investment advice, a solicitation, a research report, or a trading instruction.

Students should not use a classroom example to trade real money. A real investment decision remains the investor's responsibility and may require advice from a qualified professional.

Limits of backtest evidence

Backtest results do not represent future performance. Classroom results may be affected by information leakage, look-ahead bias, survivorship bias, underestimated transaction costs, slippage, liquidity constraints, overfitting, parameter selection, changes in market conditions, short samples, and unrealistic execution assumptions.

A high return, high Sharpe ratio, or small drawdown in class is research evidence under the stated conditions. It is not a guarantee of profit.

Permitted and prohibited uses of GenAI

GenAI may assist learning, but it may not replace the student's judgment or verification.

Permitted uses include:

- drafting a strategy hypothesis;
- explaining a programming error message;
- organizing a feature dictionary;
- revising overconfident wording; and
- generating questions or a list of possible risks for later verification.

Prohibited uses include:

- submitting GenAI output as a final answer without review;
- concealing material GenAI use;
- fabricating data, performance, execution records, or literature;
- presenting generated investment advice as reliable; and
- using generated code or a financial explanation without checking it.

The student or team remains responsible for every submitted calculation, source, line of code, interpretation, and oral answer. Do not enter credentials, personal data, restricted data, or another student's work into a GenAI service.

Required disclosure

If GenAI affects a submitted report, code file, figure, table, or presentation, include:

Tool used:

Purpose:

Main prompt:

Content retained:

Content removed or changed:

Human verification performed:

The disclosure is not a substitute for verification. During oral questions, every student must be able to explain the submitted work without asking GenAI to answer in real time.

Language and responsibility

Avoid claims such as:

This strategy guarantees profit.

The model proves this market pattern.

The AI found the best strategy.

This result will continue in the future.

Use statements whose strength matches the evidence, for example:

The evidence in this sample suggests ____.

The result is conditional on ____.

The strategy remains fragile when ____.

Before deployment, we need ____.

Minimum risk disclosure

Every strategy report must identify:

- the data source and period;
- the target and holding period;
- the benchmark;
- transaction-cost assumptions;
- maximum drawdown or another relevant downside-risk measure;
- whether information leakage was checked;
- whether cost or threshold sensitivity was examined;
- the parts affected by GenAI; and
- why the classroom evidence is insufficient for direct real-money trading.

The goal is not to find a strategy that is certain to earn a profit. The goal is to formulate a hypothesis, construct data and a model, check bias, backtest and stress test, disclose limitations honestly, and make a decision whose reasoning can be traced to evidence.

中文版本

教育用途聲明

本課程所有 notebook、範例、策略、回測與討論皆僅供教育用途，不構成任何投資建議、投資招攬、研究報告或交易指示。

學生不應根據課堂範例直接進行真實資金交易。任何真實投資決策皆需自行承擔風險，並諮詢合格專業人士。

回測限制聲明

回測結果不代表未來績效。課堂回測可能受到下列因素影響：

- data leakage
- look-ahead bias
- survivorship bias
- transaction cost underestimation
- slippage and liquidity constraints
- overfitting
- parameter selection bias
- regime change
- short sample period
- unrealistic execution assumptions

因此，課堂上看到的高報酬、高 Sharpe 或低 drawdown，都只能視為研究 evidence，不能視為獲利保證。

GenAI 使用原則

本課程允許使用 GenAI，但 GenAI 的角色是輔助，不是替代判斷。

允許用途：

- 產生策略假說草稿。
- 解釋程式錯誤訊息。
- 協助整理 feature dictionary。
- 幫助改寫過度自信的文字。
- 產生質詢問題或風險清單。

不允許用途：

- 直接把 GenAI 文字當成最終答案。
- 隱藏 GenAI 使用。
- 讓 GenAI 編造不存在的資料、績效或文獻。
- 用 GenAI 生成投資建議並宣稱可靠。
- 未經檢查就採用 GenAI 產生的程式碼或金融解釋。

學生或團隊仍須對每一項提交的計算、來源、程式碼、解釋與口頭回答負責。不得將帳號憑證、個人資料、受限制資料或其他學生的作品輸入 GenAI 服務。

GenAI 使用揭露格式

若 GenAI 影響提交的報告、程式碼、圖、表或簡報，需附上：

使用工具：

使用目的：

主要 prompt：

採納內容：

刪除或修改內容：

人工檢查方式：

揭露不能取代人工查證。口頭問答時，每位學生都必須自行說明提交內容，不得即時要求 GenAI 代答。

語言與責任

課程提案與簡報中應避免下列語句：

This strategy guarantees profit.
The model proves this market pattern.
The AI found the best strategy.
This result will continue in the future.

建議改成：

The evidence in this sample suggests _____.
The result is conditional on _____.
The strategy remains fragile when _____.
Before deployment, we need _____.

最低風險揭露要求

每份策略提案至少要揭露：

- 資料來源與時間範圍。
- target 與 holding period。
- benchmark。
- transaction cost 假設。
- 最大回撤或 downside risk。
- 是否做過 leakage 檢查。
- 是否做過成本或 threshold sensitivity。
- 使用 GenAI 的部分。
- 不建議直接真實交易的原因或條件。

課堂共識

這門課的目標不是找出「一定賺錢」的策略，而是訓練學生建立一種負責任的金融 AI 判斷能力：

提出假說

- > 建立資料與模型
- > 檢查偏誤
- > 回測與壓力測試
- > 誠實揭露限制
- > 用 evidence 做決策